Jasper Sands

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Quantum algorithms, error correction, and machine learning for qubit control. MS in computer science at Columbia. Thesis at Diraq and UNSW Sydney on automating silicon spin qubit measurement.

EDUCATION

Columbia University, MS in Computer Science, Quantum Computing Track · GPA 3.95/4.0 — Sep 2025 – May 2027

Thesis: automating silicon spin qubit measurement with machine learning. Diraq and UNSW Sydney, Sep 2026 – May 2027, co-supervised by Dr. Nard Dumoulin Stuyck and Dr. Henry Yuen.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & ML.
TA: Introduction to Quantum Computing.

Washington University in St. Louis, BS in Computer Science and Economics · GPA 3.81/4.0 — Aug 2021 – May 2025

Coursework: Machine Learning · Analysis of Algorithms · Computer Security. **TA:** Malware Analysis; Parallel Programming.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for financial Value at Risk · *CUDA-Q, C++* — Overall Winner, iQuHACK 2026

- Wrote the GPU-accelerated state preparation, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation.
- Measured query scaling matched the $O(1/\varepsilon)$ bound against an $O(1/\varepsilon^2)$ quasi-Monte Carlo classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* — qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XZZX surface codes, with exact MWPM and Union-Find decoding running client-side.
- Finite-size scaling over d = 3, 5, 7 at 20k shots/point puts the phenomenological threshold at 3.2%.

Q-Search: proof-gated quantum algorithm research and verification engine — qsearch.jaspersands.com

- Automated search for structural quantum advantage on non-abelian HSP, dihedral coset, and linear code equivalence problems.
- Formal proof gates test each mechanism against 1,200+ classical baselines across 720+ theorem modules, indexing negative results.

LLM-Based Lossless Text Compression · *PyTorch, CUDA* — Systems optimization

- Arithmetic coding on autoregressive LLM predictions: 5–16× average and 39× best-case compression, beating zstd and gzip; static KV caching and CUDA graphs for throughput.

RESEARCH

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* — Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and tuning for precision against a fixed false-positive budget.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* — May – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian by grounding outputs in molecular graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher* — Sep 2024 – Jun 2025

- Fine-tuned LLaMA-3 with Unsloth and Hugging Face for QA over 50k foster-care policy documents, with human-in-the-loop feedback.

Also: Desdr Open Insurance Toolkit, Columbia (Dr. Eugene Wu, Sep–Dec 2025): survey data into actuarial and climate-risk features.
Computer Vision & AI Lab, WashU (Dr. Umar Iqbal, Sep 2024–Jun 2025): pipelines parsing billions of forum posts.

EXPERIENCE

Smack Technologies · *Software Engineer Intern* — May – Aug 2026

- Wrote physics simulations and decision models for the core technology pipelines, plus benchmarks validating the learned policies.

Highnote · *Cybersecurity Engineer Intern* — May – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day, and wrote 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern* — May – Jul 2023

- Built internal admin and secure-query tooling in JavaScript and Rails via Retool, saving roughly 250 engineering hours a month.

Earlier: Bugcrowd, Associate Application Security Engineer (May–Jul 2020): vulnerability triage and client disclosure work.

HONORS & LEADERSHIP

- **Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026 · Qualified, NYU Abu Dhabi Quantum Hackathon for Social Good 2027 · **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026 · **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026
- **Graduate Lead**, Columbia Quantum Algorithms Reading Group (Sep 2025–Jun 2026): ran weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms lecture notes, each with simulation checks.

SKILLS

Quantum: Qiskit and Qiskit Runtime (IBM Heron hardware), CUDA-Q, Stim, PyMatching, QuTiP · Hamiltonian simulation, phase estimation, amplitude amplification and estimation, QSVT and block encoding, LCU · surface codes, MWPM and Union-Find decoding · VQE, QAOA · randomized benchmarking, tomography, noise modeling, transpilation

Languages: Python, C++, Rust, C, MATLAB, JavaScript, TypeScript, Go, Java

ML & systems: PyTorch, CUDA, Hugging Face · Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, LaTeX

Mathematics: Linear algebra, tensor networks, probability, convex optimization, quantum complexity (BQP, QMA), query complexity